Association Between Internet Use and Unmet Need for Family Planning among Ever-Married Women in Bangladesh: Evidence from BDHS 2022

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Abstract

Objective

To examine the association between internet use and unmet need for family planning among ever-married women in Bangladesh and to assess disparities by residence, administrative division, and key socio-demographic characteristics.

Methods

This study has used cross-sectional nationally representative Bangladesh Demographic and Health Survey (BDHS) 2022 data, including 4,370 ever-married women aged 15-49 years covering urban and rural areas of Bangladesh. Stepwise survey-weighted multivariable logistic regression models were estimated to evaluate the association between internet use and unmet need for family planning. Additionally, stratified analyses were performed by place of residence, administrative division, maternal age, age at first birth, maternal education, and media exposure. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC).

Results

Overall, 41% of women showed unmet need. Internet use was significantly associated with higher odds of unmet need (aOR=1.83, 95% CI:1.52–2.20). Significant associations were observed in the Barishal, Rajshahi, and Sylhet divisions and among women with early age at first birth and lower education. The model demonstrated acceptable discrimination (AUC=0.78). Spatial findings reveal particular attention is required in Chittagong to improve equitable access to contraceptive services.

Conclusions

Internet use was associated with a higher unmet need for family planning among ever-married women in Bangladesh, with substantial regional and socio-demographic heterogeneity. These findings suggest that digital access alone may not reduce unmet need without concurrent improvements in service availability and responsiveness. Targeted family planning interventions should be prioritized in high-need eastern divisions, particularly Chittagong, to reduce regional inequalities in contraceptive access.

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Keywords: Reproductive health, Digital health, Health services accessibility, Bangladesh

1.Introduction

Family planning services are not just the foundation for women's reproductive health, but also the foundation for women's empowerment, social justice, and sustainable development [1]. Family planning services allow women and their partners to make informed choices about the number and spacing of their children, and these choices have significant implications for women's health, women's empowerment, and sustainable development [2]. Despite decades of global and national investment in improving access to family planning, millions of women worldwide need for family planning, defined as the gap between what women desire and what they actually do in terms of contraceptive use [3]. Women with unmet need are at greater risk of unintended pregnancy, poor maternal health outcomes, and negative social and economic outcomes[4].

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Globally, unmet need remains widespread. Approximately 200 million women in low- and middle-income countries wish to evade pregnancy but do not use contraception [5]. Sub-Saharan Africa displays the highest prevalence, while South and Southeast Asia have lower percentages but a large absolute number of affected women [6]. Bangladesh has made notable improvement in contraceptive coverage over the past decades; modern contraceptive use among married women improved from under 10% in the 1970s to over 60% today [7]. Yet, analysis of Bangladesh Demographic and Health Survey (BDHS) 2022 indicates that 41.8% of married women of reproductive age still have an unmet need, prominence obstinate gaps in access, autonomy, and evidence [8].

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The recent years have witnessed the emergence of a digital revolution in Bangladesh, which offers new avenues for filling the gaps. Internet access, for instance, was very low in the country; in 2011, only 7.5% of women in urban areas and 0.5% of women in rural areas used the internet. In 2014, the percentage of women in urban areas had roughly doubled, reaching 13%, while the percentage of women in rural areas remained very low, at 1.8%. In the period between 2014 and 2018, the percentage of women in Bangladesh who used the internet had risen dramatically, reaching 28% in urban areas and 6% in rural areas. In 2022, the percentage of women in rural areas who used

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the internet had risen sharply to 16.6%, while the percentage of women in urban areas had remained stable at 27.3%, thus narrowing the rural-urban divide [9, 10]. The digital revolution, therefore, offers new opportunities for women in Bangladesh. The internet, unlike traditional forms of mass media, offers interactive facilities and information, which are likely to impact health and reproductive behaviors [11].

The progress of internet access is mainly appropriate for family planning. Information blocks, misunderstandings about contraception, and inadequate exposure to health facilities subsidize to high unmet need [12]. Internet use can offer women with opportune, correct information on contraception, peer sustenance, and associated to health services, potentially plummeting unmet need and empowering women to make informed decisions [13–15]. However, the association between internet use and family planning outcomes in Bangladesh remains limited explored [16]. Existing investigations mostly focus on socio-demographic determinants of unmet need age, parity, education, residence, and media exposure while overlooking digital assignation [17]. Moreover, the consequence of internet use may fluctuate across subgroups, such as maternal age, age at first birth, maternal education, and urban-rural residence, importance a critical knowledge gap [18].

Addressing unmet need aligns with Sustainable Development Goals (SDG), particularly SDG 3 (ensuring healthy lives) and SDG 5 (achieving gender equality). Finding how digital knowledges, such as internet use, effect generative health performances can notify interferences that are more reasonable, embattled, and active [14]. This is particularly significant for women in rural or downgraded societies who archeologically have challenged inadequate contact to health evidence and facilities.

Therefore, the main goal of this study is to:

1. Assess the national association between internet use and unmet need for family planing from BDHS 2022 for reproductive aged women aged (15-49) years;
2. Examine inequalities by urban-rural residence and administrative division; and
3. Investigate stratified analysis by maternal age, age at first birth, maternal education, and media access.

By addressing these purposes, this study provides policy-relevant evidence on how digital assignation can subsidize to diminish unmet need, advancing reproductive autonomy, and enhancing equitable access to family planning evidence and services across Bangladesh.

2.Methods

2.1. Study Design and Data Source

This investigation utilized a cross-sectional study design and secondary data from the nationally representative Bangladesh Demographic and Health Survey (BDHS) 2022 executed by the National Institute of Population Research and Training (NIPORT), Ministry of Health and Family Welfare, Government of Bangladesh, with methodological support from ICF International through the USAID-funded DHS Program. This household survey provides an extensive range of signs on population, health, and nutrition indicators, which are important in the procedures of tracking and estimation of health plans and strategies in Bangladesh. The BDHS 2022 employed a two-stage stratified cluster sampling design to ensure national representativeness. In the first stage, 675 enumeration areas (EAs) as key sampling units were selected with probability relational to EA size from a sampling frame following from the 2011 Bangladesh Population and Housing Census efficient through 2017-2018 cartographical procedures. The sample involved 225 urban EAs and 450 rural EAs, importance the country's demographic distribution. In the second stage, a methodical sample of around 30 households were extract from each selected EA using an equal probability systematic selection process. This sampling method shaped a total sample of 20,250 residential households across all eight administrative divisions of Bangladesh. Data collection was implemented between October 2022 and February 2023 by using Computer-Assisted Personal Interviewing (CAPI), which simplified development data excellence through real-time commendation, skipping, and reduction of data entry errors. The data collection tool caused in extremely high response rates, with 98.6% for the household questionnaire and 97.5% for the individual women's questionnaire, screening a low level of non-response bias and a high level of data excellence. The response occurrences far better international ideals for population surveys, confirming data reliability and generalizability. The BDHS 2022 collected data from 20,065 ever-married women aged 15–49 years on numerous zones including fecundity preferences, contraceptive knowledge and use, maternal and child health services, nutritional status, domestic

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violence, women's empowerment, and household characteristics. Detailed are address in BDHS report 2022 [8].

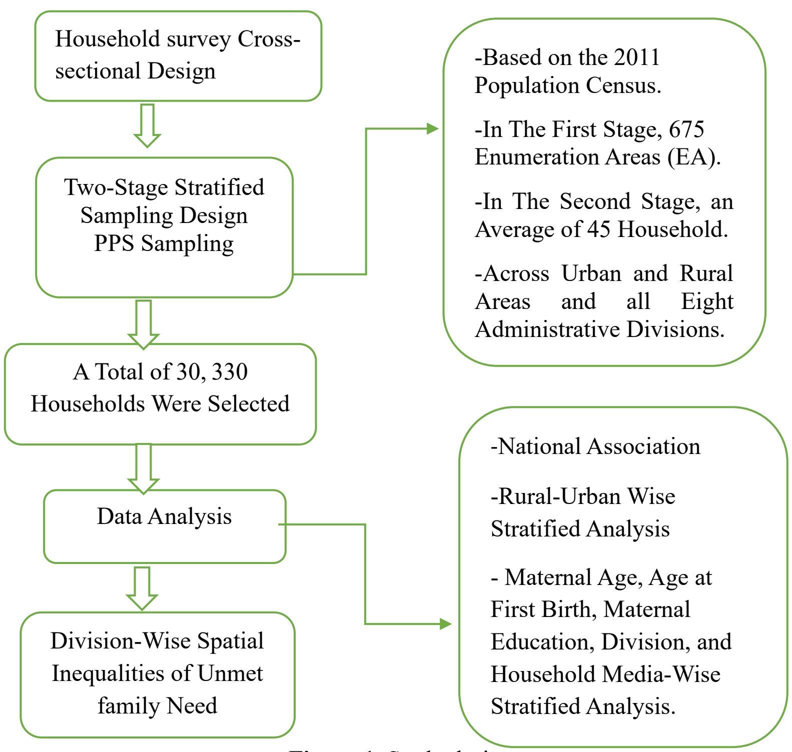

Figure 1. Study design

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2.2. Study Population and Eligibility Criteria

The target population of the study included pregnant women at the time of the survey, ranging in age from 15-49 years, who were either usual residents of the selected households or visitors present

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the night before the survey. The target population of pregnant women was selected because the unmet family planning needs of pregnant women are one of the important, yet often neglected, areas of reproductive health. While the assessment of the unmet family planning needs of women who are not pregnant is based on the use of contraceptives and fertility desires, the assessment of the unmet family planning needs of pregnant women is based on whether the pregnancy was intended, mistimed, or unwanted. The initial sample after applying the pregnancy criterion consisted of 4,370 women [**Figure 1**]. Subsequently, we excluded women with missing, ambiguous, or non-applicable responses on the outcome variable. This lucid reporting follows to STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) strategies for cross-sectional studies and allows readers to assess the representativeness and generalizability of results [19].

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2.3. Outcome Variable: Unmet Need for Family Planning

The key outcome variable was considered unmet need for family planning among currently pregnant women. This variable was extracted from the revised DHS unmet need variable, which categorizes women based on their pregnancy purpose status. Currently pregnant women were categorized as having unmet need if their current pregnancy was reported as mistimed or unwanted. The outcome was treated as a binary variable, coded as 1 for unmet need (mistimed or unwanted pregnancy) and 0 for no unmet need (intended pregnancy). Women with others excluded from these criteria. This definition follows standard DHS analytical strategies and is extensively utilized in reproductive health investigate [20].

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See the revised diagram here.

2.4. Main Exposure Variable: Internet Use

The main exposure variable was women’s internet use. Internet use was assessed based on self-reported use within the last 12 months previous to the survey. Women who defined using the internet during the past 12 months were classified as internet users, whereas those who defined never using the internet or not using it during the past 12 months were classified as non-users. This binary classification captures recent digital assignation and reflects potential exposure to online health statistics and digital communication stages.

2.5. Covariates

Covariates were selected based on theoretical relevance, prior literature, and availability within the BDHS dataset. These variables represent socio-demographic, reproductive, household, empowerment, and health-related characteristics that may confound the relationship between internet use and unmet need. Maternal age was categorized into three groups: 15–24 years, 25–34 years, and 35–49 years, reflecting distinct reproductive life stages. Age at first birth was grouped into early (10–17 years), normative (18–24 years), and late (25 years or older) categories to capture differences in reproductive timing. Age at first cohabitation was categorized as less than 18 years, 18–20 years, and 21 years or older, reflecting early marriage dynamics. Educational attainment was categorized separately for women and their husbands. Women's education was grouped as no education, primary education, and secondary or higher education. Husband's education was similarly categorized. Parity was classified into four groups: no living children, one child, two children, and three or more children. Household wealth status was derived from the DHS wealth index and categorized into poor (lowest two quintiles), middle, and rich (highest two quintiles). Place of residence was classified as urban or rural. Administrative division was included to account for regional variation. Religion was categorized as Muslim and non-Muslim. Household size was grouped into fewer than five members and five or more members. Women's employment status was classified as currently working or not working. Husband's employment status was categorized as worked versus did not work. A composite women's empowerment index was constructed using five decision-making domains: decisions regarding own health care, major household purchases, visits to relatives, control over spending money, and contraceptive decision-making. Women who participated either alone or jointly with their partner in at least three of these domains were classified as empowered, while those participating in two or fewer domains were classified as not empowered. This composite measure reflects multidimensional autonomy relevant to reproductive decision-making. A binary variable representing experience of intimate partner violence (IPV) was constructed based on women's self-reported experience of physical violence from their partner. Women reporting at least one form of physical violence were categorized as having experienced IPV. Women with missing or "don't know" responses were excluded from IPV analysis. A household media exposure variable was created to capture access to information sources. Households were classified as having media access if they owned at least one of the following: radio, television, newspaper access, or mobile phone. A household asset index was constructed based on ownership of major assets such as refrigerator, bicycle, motorcycle, or car. Housing material quality was categorized as improved or unimproved based on floor, wall, and roof materials. Ownership of house and household size were also included as indicators of socioeconomic status [21–27].

2.5. Missing Data

The proportion of missing data was minimal for most variables. For selected categorical variables with small proportions of missing values, simple mode imputation was applied to retain sample size. Sensitivity analyses indicated that imputation did not materially alter the results. For two specific covariates husband's education and husband's employment status missing data rates were slightly higher but still modest: 0.82% for husband's education (36 cases) and 1.03% for husband's employment (45 cases). For these variables, we implemented simple mode imputation, replacing missing values with the most frequently observed category in the complete cases. For husband's education, the modal category was secondary or higher education (59.50% of complete cases); for husband's employment, the mode was currently working (95.88% of complete cases).

2.6. Survey Design and Weighting

All analyses accounted for the complex survey design of BDHS 2022. Sampling weights were applied to adjust for unequal probability of selection and non-response. Primary sampling units and stratification variables were incorporated into survey-adjusted analyses to obtain nationally representative estimates and correct standard errors. This study followed STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines [28].

2.7. Model Building Stages

Step 1: Bivariate Analysis

First, the study employed Chi-square tests for assessing initial association between the covariates and the outcome variable. The categorical variables with a p-value less than 0.25 were considered for the final model for adjustment, i.e., the inclusion of the covariates in the final model was considered to avoid the exclusion of potentially important confounding variables.

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Step 2: Forward Stepwise Model Selection

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In the second step, all the potential confounding variables as covariates were added individually to the model. The inclusion of the covariates in the model was considered on the basis of statistical and conceptual considerations. The inclusion of the covariates in the model was considered on the basis of:

- Statistical Significance (p-values),
- Change in the Odds Ratio (OR) for the main exposure variable,
- Improvement in the model fit, i.e., Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).

Conceptually consistent confounding variables were retained in the model, even if they slightly worsened the AIC/BIC, in order to avoid model misspecification. The current study also employed the likelihood ratio tests (LRT) for comparing the logistic regression models, which compare to nested logistic regression models, assessing the consequence of covariates to model performances.

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2.8. Statistical Analysis

Descriptive statistics were used to summarize sample characteristics. Weighted proportions and 95% confidence intervals were calculated for unmet need and key independent variables.

Bivariate associations between unmet need and independent variables were assessed using design-adjusted chi-square tests. Variables with statistical significance at $p < 0.20$ or those considered conceptually important were included in multivariable analysis. Survey-weighted multivariable logistic regression models were fitted to examine the association between internet use and unmet need for family planning. Adjusted odds ratios (AORs) and 95% confidence intervals were reported. Model selection was guided by theoretical considerations and statistical criteria, including Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and likelihood ratio tests. Lower AIC and BIC values indicated better model fit. In this analysis, among 14 models the m13 indicates the best adjusted model with lower AIC=4885.293, and BIC= 5057.621. Stratified analyses were conducted by place of residence and maternal age to explore potential effect modification. Multicollinearity was assessed using variance inflation factors. Model goodness-of-fit was evaluated using the Hosmer–Lemeshow test. All statistical analyses were performed using Stata software. Statistical significance was set at $p < 0.05$.

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2.9. Model Diagnostics and Assumption Checking

Multiple diagnostic strategies were conducted to evaluate best adjusted model competence, find significant remarks, and confirm that assumptions essential for effective logistic regression extrapolation were contented.

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Goodness-of-fit testing was implemented using the Hosmer-Lemeshow test, which assesses how well the model's predicted probabilities match observed outcome frequencies across deciles of predicted risk. The test splits observations into groups based on predicted probability deciles and compares observed versus expected counts of outcomes in each group using a chi-square statistic. Non-significant results ($p > 0.05$) indicate acceptable model fit, while significant results suggest poor calibration. Additionally, model performance was evaluated using the ROC (Receiver Operating Characteristic) curve, with the area under the curve (AUC) indicating goodness of fit

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Multicollinearity assessment was conducted by calculating variance inflation factors (VIF) for all covariates in the final model. In practice, most covariates showed $VIF < 5$, indicating no concerning multicollinearity. In this study, the VIF of best adjusted model was 2.42 only.

2.10. Ethics Approval and Consent to Participate

The BDHS 2022 received ethical approval from the Bangladesh Medical Research Council and the ICF Institutional Review Board. The present study used publicly available, de-identified secondary data. Therefore, no additional ethical approval was required. However, details at: https://dhsprogram.com/Methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm.

2.11. Adjusted Confounders in Final Models

These set of covariates were added in final models including maternal age, age at first birth, maternal education status, husband education status, parity, place of residence, household media, household assets, husband employment, household division, women's empowerment, household religions, and contraception status.

3.Results

3.1. Background Characteristics

Overall, 41% of women revealed unmet need for family planning [**Figure 2**].

[**Insert Figure 2 Here**]

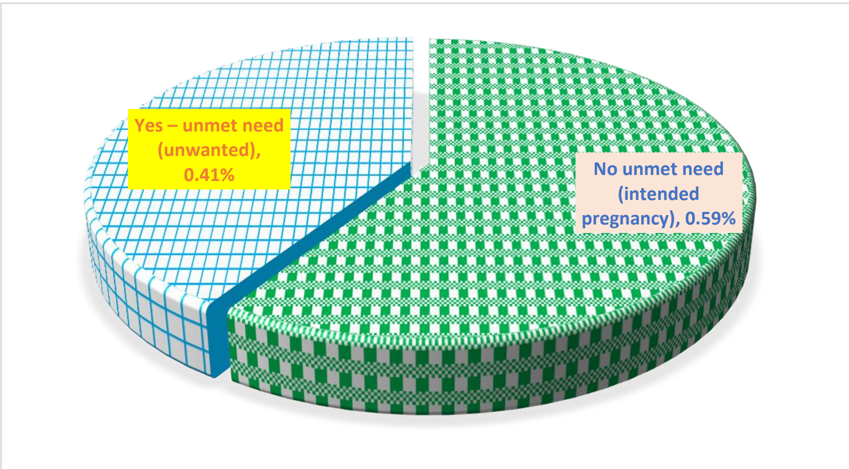

Figure 2: Prevalences of Unmet Need family Planning

Unmet need was significantly higher among older women, increasing from 13.1% in those aged 15–24 to 17.1% in 25–34 and 11.0% in 35–49 years (p<0.001), and among women who cohabited early (26.5%) compared with later cohabitation (4.6%, p<0.001). Women who have higher parity (≥3 children), showed higher (11.7%), unmet need compared to women with no children (3.96%, p<0.001). Additionally, women with no education showed higher unmet need (3.4%) compared to above secondary education (28.4%, p<0.001). Early age at first birth was associated with higher unmet need (13.9% vs 6.3% in late first births, p<0.001). Women not participate in joint contraceptive decisions (10.2%) and those lacking maternal empowerment (10.8%) experienced higher unmet need (p<0.001 and p=0.001, respectively). Husband's lower education (p<0.001) and non-working status (p=0.007), female-headed households (14.99% vs 26.27% in male-headed, p<0.001), and households with unimproved materials (4.87%, p=0.001) or no assets (14.71%, p=0.040) were also significantly associated. Media access (36.2% vs 5.1%, p<0.001) and internet use (p<0.001) were associated to lower unmet need [**Table 1**].

[**Insert Table 1 Here**]

Table 1. Distribution of unmet need for family planning by background characteristics (N=4,370)

Unmet Need for Family Planning				
Characteristics	**No n (%)**	**Yes n (%)**	**Total n (%)**	**χ² Test, (p-value)**
Demographic Characteristics				
Women Age (years)				297.4116 (**<0.001**)
15- 24	1,317(30.14)	573(13.11)	1,890(43.25)	
25-34	1,019 (23.32)	749 (17.14)	1,768(40.46)	
35-49	231(5.29)	481(11.01)	712(16.29)	
Age at Cohabitations				47.0128 (**<0.001**)
10–17 (Early)	1,411(32.29)	1,157(26.48)	2,568(58.76)	
18-24(Mild)	706(16.16)	444(10.16)	1,150(26.32)	
≥25 (Late)	450(10.30)	202(4.62)	652(14.92)	
Parity				854.1223(**<0.001**)
No Children	990(22.65)	173(3.96)	1,163(26.61)	
One Children	955(21.85)	479(10.96)	1,434(32.81)	
Two Children	478(10.94)	639(14.62)	1,117(25.56)	
≥3 Children	144 (3.30)	512(11.72)	656(15.01)	
Household Division				31.6240 (**<0.001**)
Barishal	246(5.63)	187(4.28)	433(9.91)	
Chittagong	456(10.43)	408(9.34)	864(19.77)	
Dhaka	403(9.22)	288(6.59)	691(15.81)	
Khulna	316(7.23)	199(4.55)	515(11.78)	
Mymensingh	279)6.38)	162(3.71)	441(10.09)	
Rajshahi	284(6.50)	150(3.43)	434(9.93)	
Rangpur	264(6.04)	156(3.57)	420(9.61)	
Sylhet	319(7.30)	253(5.79)	572(13.09)	
Place of Residence				10.0405 (0.002)
Urban	852(19.50)	517(11.83)	1,369(31.33)	
Rural	1,715(39.24)	1,286(29.43)	3,001968.67)	
Household Religion				0.65 (0.723)
Non-Muslim	215(4.92)	112(2.56)	327(7.48)	
Muslim	2,352(53.82)	1,691 (38.70)	4,043(92.52)	
Maternal Characteristics				
Maternal Education				71.4741(**<0.001**)
Not education	102(2.33)	150(3.43)	252(5.77)	
Primary Complete	429(9.82)	413(9.45)	842(19.27)	
Above Secondary	2,036(46.59)	1,240(28.38)	3,276(74.97)	
Maternal Employment				0.0267 (0.870)
Not Working	2,025(46.34)	1,426(32.63)	3,451(78.97)	
Currently working	542(12.40)	377(8.63)	919(21.03)	
Age at first Birth				413.5423(**<0.001**)

(10-17) Early	478 (10.94)	609(13.94)	1,087(24.87)	
(18-24) Normal	964 (22.06)	920(21.05)	1,884(43.11)	
(≥25) Late	1,125(25.74)	274(6.27)	1,399(32.01)	
Contraception Decision				
Not joint	459(10.50)	444(10.16)	903(20.66)	29.3903 (**<0.001**)
Joint	2,108(48.24)	1,359(31.10)	3,467(79.34)	
Maternal Empowerment				
No	784(17.94)	470(10.76)	1,254(28.70)	10.3601 (0.001)
Yes	1,783(40.80)	1,333(30.50)	3,116(71.30)	
Husband Characteristics				
Husband Education				27.9075 (**<0.001**)
No Education	297(6.80)	282(6.45)	579(13.25)	
Primary Complete	642(14.69)	513(11.74)	1,155(26.43)	
Above Secondary	1,628(37.25)	1,008(23.07)	2,636(60.32)	
Husband Employment				7.3836(0.007)
Not Working	64(1.46)	71(1.62)	135(3.09)	
Working	2,503(57.28)	1,732(39.63)	4,235(96.91)	
Household Head Sex				279.7224 (**<0.001**)
Male	2,194(50.21)	1,148(26.27)	3,342 (76.48)	
Female	373(8.54)	655(14.99)	1,028(23.52)	
Household & Socioeconomic Factors				
Household Wealth Status				1.0750(0.584)
Poor	880(20.14)	635(14.53)	1,515(34.67)	
Middle	518(11.85)	342(7.83)	860(19.68)	
Rich	1,169(26.75)	826(18.90)	1,995(45.65)	
Household Size				1.0537(0.305)
Less than 5	1,195(27.35)	1,372(21.40)	2,006(45.90)	
≥ 5	811(18.56)	992(22.70)	2,364(54.10)	
Household Ownership				2.2373(0.135)
Does not Own	2,458(56.25)	1,209(39.11)	4,167(95.35)	
Owns	109(2.49)	94(2.15)	203(4.65)	
Household Assets				4.2316 (0.040)
No	994(22.75)	643(14.71)	1,637(37.46)	
Yes	1,573(36.00)	1,160(26.54)	2,733(62.54)	
Household Materials				11.2949 (0.001)
Unimproved	395(9.04)	213(4.87)	608(13.91)	
Improved	2,172(49.70)	1,590(36.38)	3,762(86.09)	
Media & Information Access				
Household Media Access				14.7085(**<0.001**)
No Media	425 (9.73)	223(5.10)	648(14.83)	
Yes Media	2,142(49.02)	1,580(36.16)	3,722(85.17)	
Internet Use Status				**22.8146 (<0.001)**
No	7714 (98.61)	109 (1.39)	2,345(53.660	
Yes	6030 (98.40)	98 (1.60)	2,025(46.34)	

3.2. Association between Internet Use and Unmet Need for Family Planning

At the national level, women who reported using the internet in the past 12 months had significantly higher odds of unmet need for family planning compared with non-users (adjusted odds ratio [aOR] 1.83, 95% CI 1.52–2.20; p<0.001). Crude estimates showed a similar pattern (cOR 1.34, 95% CI 1.18–2.14; p=0.003). Stratified analyses revealed consistent associations across rural (aOR 1.86, 95% CI 1.50–2.31; p<0.001) and urban (aOR 1.85, 95% CI 1.28–2.70; p=0.001) settings. All models were adjusted for potential confounders, including maternal age, age at first birth, parity, husband's education and occupation, household assets, media access, women's empowerment, place of residence, religion, and contraception status. These findings indicate that internet use is independently associated with higher unmet need for family planning, with similar effects in rural and urban populations [**Table 2**].

[**Insert Table 2 Here**]

Table 2: Associations Between Unmet *Need for Family Planning and Internet Use Status* (N= 4,370, N1 Rural= 3,001, N2 Urban = 1,369)

Characteristics	**Outcome Variable: Unmet Need for Family Planning**			
Main Exposure: Internet Use Status	**cOR (95% CI) p**	**aOR (95% CI) p**		
	-	**National Level**	**Rural**	**Urban**
Non-users of internet in the last 12 months (Ref)	1.00	1.00	1.00	1.00
Internet use last 12 months	1.34(1.18-2.14) 0.003	1.83(1.52-2.20), **<0.001**	1.86 (1.50-2.31), **<0.001**	1.85 (1.28-2.70), 0.001

Commented [A24]: Is this for national/rural/urban??? Why do you have 1 value in the cOR?

Note: cOR = Crude Odds Ratio; aOR= Adjusted Odds Ratios CI = confidence interval

Confounders: maternal age, age at first birth, husband education, parity, place of residence, access of household media, household assets status, husband occupation, women's empowerment, household religion, contraception status

3.3. Stratified Association between Internet Use and Unmet Need for Family Planning

Table 3: Maternal Age, Age at First Birth, Division, Maternal Education, and Household Media-wise Associations Between Internet Use and Unmet Need for Family

Commented [A25]: Is this from the same analysis as table 2?

Variable	aOR (95% CI)
Age at First Birth	
Early (10–17)	2.70 (1.91–3.82)
Normal (18–24)	1.67 (1.29–2.17)
Late (≥25)	1.38 (0.95–1.99)
Age Group (years)	
15–24	1.42 (1.09–1.85)
25–34	2.41 (1.82–3.17)
35–49	2.30 (1.40–3.75)
Division	
Barishal	3.24 (1.99–5.28)
Chattogram	1.98 (1.27–3.08)
Dhaka	1.63 (1.08–2.46)
Khulna	1.47 (0.89–2.42)
Mymensingh	1.50 (0.80–2.79)
Rajshahi	3.30 (1.62–6.72)
Rangpur	0.86 (0.47–1.55)
Sylhet	2.98 (1.96–4.53)
Maternal Education	
No Education	5.34 (1.70–16.71)
Primary Education	3.08 (2.01–4.72)
Secondary and Higher	1.62 (1.32–2.00)
Household Media Exposure	
No Media	1.89 (0.97–3.67)
Has Media	1.85 (1.51–2.24)

Note: *Odds ratios (ORs) compare internet users with non-users (reference group, OR=1.0). ORs >1 indicate higher odds of unmet need among users. 95% confidence intervals are shown; CIs excluding 1 denote statistically significant associations*.

Women with early (10–17 years) and normal (18–24 years) age at first birth showed higher odds compared with those with late first birth (aOR=2.70, 95% CI: 1.91–3.82; OR=1.67, 95% CI: 1.29–2.17). Likewise, by age group, internet users aged 15–24, 25–34, and 35–49 years observed significantly higher odds (aOR=1.42–2.41) than non-users. Geographically, significant associations were observed in Barishal (aOR=3.24, 95% CI: 1.99–5.28), Chattogram (aOR=1.98, 95% CI: 1.27–3.08), Dhaka (aOR=1.63, 95% CI: 1.08–2.46), Rajshahi (OR=3.30, 95% CI: 1.62–6.72), and Sylhet (aOR=2.98, 95% CI: 1.96–4.53). Regarding maternal education, women with no education (aOR=5.34, 95% CI:1.70–16.71), primary (aOR=3.08, 95% CI: 2.01–4.72), and secondary or higher education (aOR=1.62, 95% CI: 1.32–2.00) all showed significantly higher odds compared with no internet users. Finally, household media exposure was associated with higher odds among those with media access (aOR=1.85, 95% CI: 1.51–2.24) [**Table 3**].

[**Insert Figure 3 Here**]

The ROC analysis confirmed acceptable discriminatory performance of the model, with an area under the curve (AUC) of 0.7783. This shows that the model properly discriminates between outcome groups approximately 78% of the time, highlighting moderate-to-good predictive accuracy and performance considerably better than chance (AUC=0.5) [**Figure 3**].

[**Insert Figure 3**]

3.4. Spatial Inequality of Unmet Family Need

Unmet need is maximum in the Chittagong (~9%) and minimum in the western divisions such as Rajshahi, Rangpur, Khulna, and Barisal (~3–4%). This east–west gradient reveals persistent regional inequalities in contraceptive utilization, with programmatic attention needed in high-need eastern divisions [**Figure 4**].

[**Insert Figure 4 Here**]

4.Discussion

This study studied the association between internet use and unmet need for family planning among ever-married women in Bangladesh. Mostly, three key findings observed. First, internet use was significantly associated with higher odds of unmet need for family planning at the national level, even after adjustment for demographic, socioeconomic, empowerment, and household factors. Second, the association insisted across rural and urban settings, representing that the observed association is not constrained to a particular residential context. Third, substantial heterogeneity was acknowledged across administrative divisions and socio-demographic strata such as regional inequalities in unmet need for family planning, with Chittagong showing the highest levels and the western divisions exhibiting the lowest. The association was particularly strong in Barishal, Rajshahi, and Sylhet, among women with early age at first birth, and among those with lower educational attainment. The model demonstrated acceptable discriminatory performance (AUC=0.78), supporting moderate-to-good predictive accuracy. Finally, these findings show that digital assignation in Bangladesh is associated with higher unmet need in several contexts rather than uniformly reducing reproductive health gaps.

While digital access is often implicit to progress reproductive health outcomes, our findings suggest a more complex and context-dependent association[29] . Internet use may growth exposure to contraceptive knowledge, fertility planning ideas, and reproductive truths dissertation without ensuring parallel progresses in facility availability [30]. In such surroundings, women may change clearer partialities to postponement or limit childbearing but encounter persistent structural blocks including stock-outs, cost barriers, provider bias, limited method choice, or sociocultural struggle that avoid execution of those likings [31].

Thus, internet use may not growth unmet need, but rather reveal formerly articulated or latent demand for unmet family need [20, 32, 33] . In this outline, higher unmet need among internet users reproduces an information access gap rather than an injurious consequence of digital exposure. Digital access may enhance intellectual and informational empowerment, growing women's autonomy in fertility decision-making [34]. However, empowerment advances may not be consistently harmonized by changes in household negotiating power or partner sustenance [35, 36]. Women exposed to online satisfied may desire smaller relatives or longer arrangement, but decision-making remains controlled by partner disagreement or gender standards [37]. This pressure between enhanced consciousness and insistent structural limitations may apparent statistically as complex unmet need among internet users [38]. The significant associations detected in Barishal, Rajshahi, and Sylhet advise geographic inequalities in digital exposure explains into service. These disparities suggest that access to and utilization of contraceptive services remain uneven across Bangladesh, underscoring the need for targeted interventions in high-need regions [39].These divisions may experience lower health system awareness, lower contraceptive service exposure, or inequalities in excellence of care [40]. Digital inequality may also play a role: internet use in some areas may be more socially restricted, less health-focused, or more partial by non-verified evidence sources [41]. The observed clustering specifies that digital development interacts with local service settings rather than functioning individualistically. Significant associations found among women aged 25-49 years align with life course theory. Women in this group may have reached or approached their desired family size. Exposure to digital content about family planning may enhance fertility preferences, hence increasing unmet needs if service provision does not match their preferences [42]. Women in the 15-24-year may have a low level of expressed demand for family planning, hence lower associations in this group [43].

Previous research from LMICs countries has generally found positive relationships between media use and contraceptive use. However, the majority of the previous research examined traditional forms of mass media, rather than internet use [44–46]. Recent research on digital health has indicated that the internet contributes to knowledge on sexual and reproductive health, especially for youth and in urban populations[47] . However, more recent research has indicated that while digital media may contribute to knowledge, it does not necessarily lead to behavior change and that there are systemic barriers [48, 49]. Our study contributes to the digital health literature by showing that internet use in Bangladesh is related to unmet need, suggesting that while knowledge may be gained, it does not necessarily lead to behavior change in the form of contraceptive use.

4.1. Strengths and Limitations

This study has several strengths such as using nationally representative BDHS 2022 data improves external rationality such as large sample size consents strong stratified analyses. Likewise, adjustment for an inclusive set of potential confounders fortifies internal rationality. Additionally, forward stepwise conceptual, and statistical methods of selecting final best model enhances robustness of methodology. Finally, several stratified analyses provide realistic insights for making evidence-based policy. ROC offers objective evaluation of model insight.

4.2. Limitations

This study with several limitations consistent with STROBE commendations such as cross-sectional design limits causal inference and averts highlighting temporalty. Self-reported internet use does not capture regularity, determination, stand type, or exposure to health-specific satisfied. Unmet need classification depends on DHS algorithm-based procedures and may be issue to reporting bias. Additionally, unavailability of potential confounders out of DHs survey data. Future longitudinal studies are needed to establish temporality between internet use and unmet need. Additionally, mixed-methods could clarify how women interpret online reproductive health information and why informational gains may not explain into contraceptive uptake.

4.3. Policy and Programmatic Implications

The findings propose that increasing digital access alone is inadequate to diminish unmet need. Policy approaches should focus on integrating authenticated family planning information on digital platforms. Strengthening the connection between online information and offline services. Increasing telehealth and digital referral services. Supporting regional differences in service quality and availability. Fostering digital literacy to address misinformation. Integrating male engagement strategies to address partner-level barriers. Regional interventions could be particularly important in divisions with larger digital unmet need associations.

Exploring interaction effects between internet use and empowerment, partner attitudes, and service quality would also deepen understanding.

4.4. Conclusions

In this nationally representative investigation, internet use was significantly associated with higher unmet need for family planning among ever-married women in Bangladesh, with considerable regional and socio-demographic heterogeneity. These findings propose that digital engagement alone does not assurance enhanced reproductive health outcomes. Bridging the gap between online information and accessible, high-quality contraceptive services is essential to translate digital expansion into meaningful reductions in unmet need and enhanced reproductive autonomy. Policymakers should prioritize targeted family planning programs in high-need eastern divisions, particularly Chittagong, to address regional disparities.

4.5. List of Abbreviations

Abbreviation	Full Form
AOR	Adjusted Odds Ratio
cOR	Crude Odds Ratio
OR	Odds Ratio
CI	Confidence Interval
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
BDHS	Bangladesh Demographic and Health Survey
DHS	Demographic and Health Survey
FP	Family Planning
UN	Unmet Need
LMICs	Low- and Middle-Income Countries
WHO	World Health Organization
SDGs	Sustainable Development Goals
BMI	Body Mass Index (if used)
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology

4.6. Declaration

4.7. Availability of data statement:

This secondary dataset is available on request from DHS program website at:

(https://www.dhsprogram.com/data).

4.8. Code Availability

The code supporting the findings of this study is available at this repository :

https://github.com/muhammadsalek/BDHS-Unmet-Need-Analysis/blob/main/README.md

4.9. Generative AI Assistance

The study used generative AI tools like Grammarly, QuillBot, and ChatGPT (GPT-4) just to improve the language, didn't use AI for any data work no generating, analyzing, or interpreting.

4.10. Consent for publication

Not applicable.

4.11. Competing interest

All of authors have no competing interests.

4.12. Funding

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4.14. Authorship Contribution Statement

Md Jamal Uddin (Corresponding Author): Investigation, Methodology, Resources, and Final Manuscript Supervision.

Md Salek Miah: Conceptualization, Data Curation, Formal Analysis, Writing - Original Draft, Resources, Project Administration.

References

1. Nazir K, Nobi Siam MM, Islam ATMS. Techno-economic feasibility and sensitivity analysis of stand-alone hybrid renewable energy system: a case study at university campus in Bangladesh. Energy Conversion and Management: X. 2026;29:101514. https://doi.org/10.1016/j.ecmx.2025.101514.

2. Ammie M, Arefaynie M, Adane B, Hussein K, Hassan M. Determinants of unmet need for family planning among currently married reproductive age women at Dewa Chefa District of Oromia special zone, Amhara region, Ethiopia, 2021; a case-control study. BMC Womens Health. 2024;24. https://doi.org/10.1186/s12905-024-02939-x.

3. Cleland J, Bernstein S, Ezeh A, Faundes A, Glasier A, Innis J, et al. Sexual and Reproductive Health 3 Family planning: the unfi nished agenda. www.thelancet.com. 2006;368:1810–37. https://doi.org/10.1016/S0140.

4. Ruiz-Rivas U, Hernández-Jiménez F, Martínez-Crespo J, Tirado-Herrero S. Unmet shelter: Extreme energy poverty, informal connections, and thermal discomfort in Cañada Real, Madrid. Energy Build. 2026;350. https://doi.org/10.1016/j.enbuild.2025.116672.

5. Singh S, Darroch JE, Ashford LS, Vlassoff M. Adding It Up: The Costs and Benefits of Investing in Family Planning and Maternal and Newborn Health.

6. PRB. The Faces of Unmet Need for Family Planning. 2012.

7. Kundu S, Kundu S, Rahman MA, Kabir H, Al Banna MH, Basu S, et al. Prevalence and determinants of contraceptive method use among Bangladeshi women of reproductive age: a multilevel multinomial analysis. BMC Public Health. 2022;22. https://doi.org/10.1186/s12889-022-14857-4.

8. Bangladesh Demographic and Health Survey 2022 Final Report. 2024.

9. Miah MS, Ullah MO. Associations of internet use and pregnancy loss with depression and anxiety among women in Bangladesh: evidence from the 2022 BDHS. BMC Womens Health. 2025;26:6. https://doi.org/10.1186/s12905-025-04166-4.

10. Zahid A, Yadav UN, Mistry SK. Digital health services in Bangladesh—The need for a sustainable design framework. World Medical and Health Policy. 2023;15:682–95. https://doi.org/10.1002/wmh3.580.

11. Pham-Thi T, Wong P. Internet Access and Youth's Mental Health and Well-being: Evidence from Ethiopia. Maastricht; 2025. https://doi.org/10.53330/GYMA8538.

12. Pickard A, Islam MI, Ahmed MS, Martiniuk A. Role of internet use, mobile phone, media exposure and domestic migration on reproductive health service use in Bangladeshi married adolescents and young women. PLOS Global Public Health. 2024;4. https://doi.org/10.1371/journal.pgph.0002518.

13. Alam MA, Sajib MRUZ, Rahman F, Ether S, Hanson M, Sayeed A, et al. Implications of Big Data Analytics, AI, Machine Learning, and Deep Learning in the Health Care System of Bangladesh: Scoping Review. Journal of Medical Internet Research. 2024;26. https://doi.org/10.2196/54710.

14. Shabuz ZR, Haque ME, Islam MK, Bari W. Link between unmet need and economic status in Bangladesh: gap in urban and rural areas. BMC Womens Health. 2022;22. https://doi.org/10.1186/s12905-022-01752-8.

15. Khan MHR, Siddik AB, Islam T. Disparities in contraceptive preferences among Bangladeshi women: a multilevel logistic regression study. BMC Public Health. 2025;25. https://doi.org/10.1186/s12889-025-24424-2.

16. Wani C, McCann L, Lennon M, Radu C. Digital Mental Health Interventions for Adolescents in Low- and Middle-Income Countries: Scoping Review. Journal of Medical Internet Research. 2024;26. https://doi.org/10.2196/51376.

17. Khan MN, Harris ML, Huda MN, Loxton D. A population-level data linkage study to explore the association between health facility level factors and unintended pregnancy in Bangladesh. Sci Rep. 2022;12. https://doi.org/10.1038/s41598-022-19559-w.

18. Khan MN, Khanam SJ, Ali Khan MM, Islam MM, Harris ML. Effects of lower tiers of government healthcare facilities on unmet need for contraception and contraception use in Bangladesh. PLoS One. 2023;18 8 AUGUST. https://doi.org/10.1371/journal.pone.0290468.

19. Vandenbroucke JP, Von Elm E, Altman DG, Gøtzsche PC, Mulrow CD, Pocock SJ, et al. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): Explanation and Elaboration. PLoS Medicine | www. 2007;4. https://doi.org/10.1371/journal.pmed.

20. Endeshew DS, Workie DL, Mitku AA. Spatial distribution and its associated factors of unmet need for family planning among women of reproductive age in administrative woredas of Ethiopia. BMC Womens Health. 2026. https://doi.org/10.1186/s12905-026-04283-8.

21. Negash WD, Wubneh HD. Skilled birth attendance and its associated factors in Chad and Nigeria: A multilevel analysis of DHS data. PLOS Global Public Health. 2025;5:e0005290. https://doi.org/10.1371/journal.pgph.0005290.

22. Gunarathne L, Nedeljkovic M, Apputhurai P, Bhowmik J. Impact of Intimate Partner Violence on mental health among married women in Sri Lanka: a study based on Women's

Wellbeing Survey-2019. Journal of Public Health (United Kingdom). 2024;46:e410–8. https://doi.org/10.1093/pubmed/fdae082.

23. Mugo NS, Agho KE, Dibley MJ. Risk Factors for Non-use of Skilled Birth Attendants: Analysis of South Sudan Household Survey, 2010. Matern Child Health J. 2016;20:1266–79. https://doi.org/10.1007/s10995-016-1928-x.

24. Palamuleni ME. Levels, Trends and Determinants of Skilled care during delivery in Malawi: Evidence from Demographic and Health Surveys. 1992.

25. Tessema ZT, Tesema GA. Pooled prevalence and determinants of skilled birth attendant delivery in East Africa countries: a multilevel analysis of Demographic and Health Surveys. Ital J Pediatr. 2020;46. https://doi.org/10.1186/s13052-020-00943-z.

26. Haq I, Sakib S, Talukder A. Sociodemographic Factors on Contraceptive Use among Ever-Married Women of Reproductive Age: Evidence from Three Demographic and Health Surveys in Bangladesh. Med Sci (Basel). 2017;5. https://doi.org/10.3390/medsci5040031.

27. Tessema ZT, Worku MG, Tesema GA, Alamneh TS, Teshale AB, Yeshaw Y, et al. Determinants of accessing healthcare in Sub-Saharan Africa: A mixed-effect analysis of recent Demographic and Health Surveys from 36 countries. BMJ Open. 2022;12. https://doi.org/10.1136/bmjopen-2021-054397.

28. Vandenbroucke JP, Von Elm E, Altman DG, Gøtzsche PC, Mulrow CD, Pocock SJ, et al. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): Explanation and Elaboration. PLoS Medicine | www. 2007;4. https://doi.org/10.1371/journal.pmed.

29. Alagarajah J, Ceccolini D, Butler S. Digital mental health interventions for treating mental disorders in young people based in low-and middle-income countries: A systematic review of the literature. Global Mental Health. 2024;11. https://doi.org/10.1017/gmh.2024.71.

30. Kim J, Aryee LMD, Bang H, Prajogo S, Choi YK, Hoch JS, et al. Effectiveness of Digital Mental Health Tools to Reduce Depressive and Anxiety Symptoms in Low- and Middle-Income Countries: Systematic Review and Meta-analysis. JMIR Mental Health. 2023;10. https://doi.org/10.2196/43066.

31. Naslund JA, Aschbrenner KA, Araya R, Marsch LA, Unützer J, Patel V, et al. Digital technology for treating and preventing mental disorders in low-income and middle-income countries: a narrative review of the literature. The Lancet Psychiatry. 2017;4:486–500. https://doi.org/10.1016/S2215-0366(17)30096-2.

32. Jiaojiao H, Jiaying Y, Yuling T. Regional variations in serum pepsinogen levels and their influencing factors: a multi-center cross-sectional study. Sci Rep. 2026. https://doi.org/10.1038/s41598-026-38326-9.

33. Sichalwe MM, Basit A, Tavengana G, Kimaro RR, Behera MR, Rehman AU. Unmet need for modern contraceptives and its associated factors among postpartum adolescent girls and young women in rural Mara, Tanzania: A facility-based cross-sectional study. Sexual & Reproductive Healthcare. 2026;47:101195. https://doi.org/10.1016/j.srhc.2026.101195.

34. Smith KA, Hardy A, Vinnikova A, Blease C, Milligan L, Hidalgo-Mazzei D, et al. Digital Mental Health for Schizophrenia and Other Severe Mental Illnesses: An International Consensus on Current Challenges and Potential Solutions. JMIR Ment Health. 2024;11. https://doi.org/10.2196/57155.

35. SWEET L, O'Driscoll K, Blums T. Women's experience of an Australian metropolitan publicly funded homebirth program: Respect, integrity, trust, and empowerment (RITE). Women and Birth. 2026;39. https://doi.org/10.1016/j.wombi.2025.102151.

36. Asmare AA, Tegegne AS, Belay DB. Unveiling hidden heterogeneity and inequalities in the continuum of care for reproductive, maternal, and child health services in sub-Saharan Africa: A multilevel latent class analysis approach. Glob Epidemiol. 2026;11. https://doi.org/10.1016/j.gloepi.2025.100237.

37. Al Mamun MA, Ahmed A, Nishat MM. Student mental health and academic motivation in the age of digital connectivity: A self-determination theory perspective. Social Sciences and Humanities Open. 2025;12. https://doi.org/10.1016/j.ssaho.2025.101927.

38. Mwase C, Nkhoma K, Allsop MJ. The role of digital health in palliative care for people living with HIV in sub-Saharan Africa: A systematic review. Digit Health. 2022;8. https://doi.org/10.1177/20552076221133707.

39. Rahman MM, Ara T, Aninda MHHMK, Nurul A, Haider MM. Rural-urban differentials in the influences of individual and geospatial preparedness on institutional childbirth: a cross-sectional study in Bangladesh. BMJ Open. 2022;12. https://doi.org/10.1136/bmjopen-2021-060718.

40. Boer M, Stevens GWJM, Finkenauer C, de Looze ME, van den Eijnden RJJM. Social media use intensity, social media use problems, and mental health among adolescents: Investigating directionality and mediating processes. Comput Human Behav. 2021;116. https://doi.org/10.1016/j.chb.2020.106645.

41. Chakrabarti S. Digital psychiatry in low-and-middle-income countries: New developments and the way forward. World J Psychiatry. 2024;14:350–61. https://doi.org/10.5498/wjp.v14.i3.350.

42. Zhen Z, Tang D, Wang X, Feng Q. The impact of digital technology on health inequality: evidence from China. BMC Health Serv Res. 2024;24. https://doi.org/10.1186/s12913-024-12022-8.

43. Pham-Thi T, Wong P. Internet Access and Youth's Mental Health and Well-being: Evidence from Ethiopia. Maastricht; 2025. https://doi.org/10.53330/GYMA8538.

44. Tasnim F, Abedin S, Rahman MdM. Mediating role of perceived stress on the association between domestic violence and postpartum depression: cross-sectional study in Bangladesh. BJPsych Open. 2023;9. https://doi.org/10.1192/bjo.2022.633.

45. Vannucci A, Flannery KM, Ohannessian CMC. Social media use and anxiety in emerging adults. J Affect Disord. 2017;207:163–6. https://doi.org/10.1016/j.jad.2016.08.040.

46. Mekonen LD, Kumsa DM, Adamu Amanu A. Social media use, effects, and parental mediation among school adolescents in a developing country. Heliyon. 2024;10. https://doi.org/10.1016/j.heliyon.2024.e27855.

47. Zhen Z, Tang D, Wang X, Feng Q. The impact of digital technology on health inequality: evidence from China. BMC Health Serv Res. 2024;24. https://doi.org/10.1186/s12913-024-12022-8.

48. Adedinsewo D, Eberly L, Sokumbi O, Rodriguez JA, Patten CA, Brewer LPC. Health Disparities, Clinical Trials, and the Digital Divide. Mayo Clinic Proceedings. 2023;98:1875–87. https://doi.org/10.1016/j.mayocp.2023.05.003.

49. Onu JU, Onyeka TC. Digital psychiatry in Nigeria: A scoping review. South African Journal of Psychiatry. 2024;30. https://doi.org/10.4102/sajpsychiatry.v30i0.2115.

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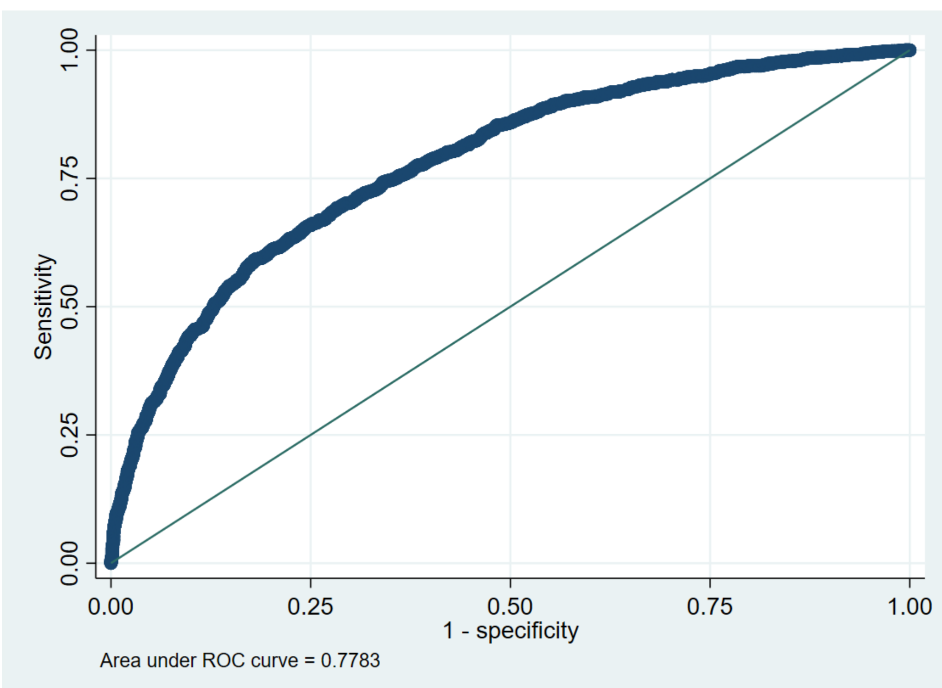

Figure 3: Receiver Operating Characteristic (ROC) Curve for the Prediction Model of Unmet Need for Family Planning in Bangladesh (BDHS 2022)

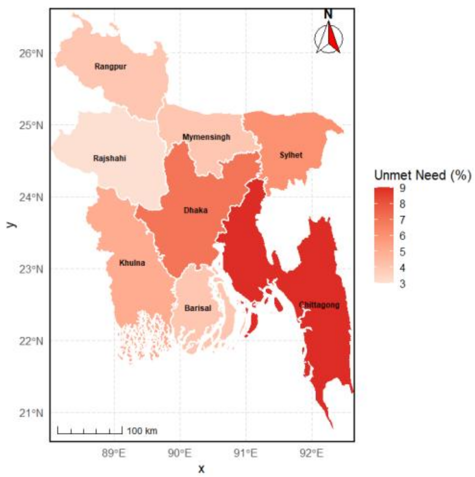

Figure 4: Spatial Inequalities of Unmet Family Need